

REACHING FOR THE SKIES
Vikram-S lifts off at the ISRO launch centre in Sriharikota, Nov. 18

Pawan Chandana looked tense as he sat glued to the computer monitor at the mission control room waiting for the countdown for the launch of the Vikram-S rocket to end on November 18. Next to him sat the reticent Bharat Daka, his partner in founding Skyroot, a start-up venture they set up in 2018 to build spacecraft. Behind them in the visitor's gallery was seated a galaxy of distinguished space scientists, including S. Somanath, chairperson of the Indian Space Research Organisation (ISRO). That Jitendra Singh, the Union minister of state for space and atomic energy, was present too was evidence of the importance of the mission called Prarambh, or the beginning. Indeed, Vikram-S was to be the first launch vehicle for spacecraft entirely designed and built by an Indian private company and whose success was expected to mark a giant leap for the country's private sector into space.

Outside, it was a balmy day at the ISRO launch centre in Sriharikota in coastal Andhra Pradesh. The six-metre-tall deep blue and white Vikram-S strapped to steel girders for the launch looked small compared to the giant launch vehicles that ISRO builds. But given the complexities of space technology, it was a huge challenge for a start-up. Chandana and Daka, both in their early 30s, were up for it. Both ex-IITians—Chandana from Kharagpur and Daka from Madras—they had chosen to join ISRO in 2012 rather than go abroad after graduation. Having worked in rocketry and other space activities sections at ISRO for six years, they decided to set up Skyroot as a private venture.

In doing so, they drew inspiration from the likes of billionaire entrepreneur Elon Musk of Tesla whose company Space X has built rockets to launch satellites and even astronauts for the National Aeronautics and Space Administration (NASA), the US government space agency. Then there was Jeff Bezos, another billionaire and owner of Amazon, who set up Blue Origin, and Richard Branson, owner of Virgin Airlines, who formed Virgin Galactic. In addition to doing what Musk does at SpaceX, both Bezos and Branson also pioneered space tourism to enable wealthy civilians to get a taste of orbiting in space and experience zero gravity or weightlessness for a payment.

When Chandana and Daka set out to build rockets four years ago, they thought government policy would be their biggest hurdle. Ever since ISRO was formed in 1969, the government had given it the monopoly over all space activity in the country. ISRO surprisingly did the job well and has a record of achievements few public sector institutions in India can rival. It propelled India into the small but exclusive club of space-faring nations who can build sophisticated heavy-lift rockets and launch a range of complex satellites designed for remote sensing, weather forecasting, telecommunications and military applications. ISRO has also conducted space exploration missions, sending probes, orbiters and landers to the Moon and Mars. To its credit, ISRO had also started steadily outsourcing the building of its launch vehicles and satellite systems to private players. As a result, both private and public sector companies now provide over 50 per cent of the country's rocket and satellite systems.

However, while ISRO made India self-reliant in much of space technology, its stranglehold over all space activity in the country had begun to choke growth. Internationally, the space sector, including building rockets and satellites, has grown to a lucrative $440 billion. But India accounts for just $8 billion, or less than two per cent, of it. To become a global space power, India needs humongous amounts of money, which the government, given the budget constraints, has found difficult to provide. They, therefore, decided to enlist the private sector in a big way, as is the case in developed countries, and provide incentive for both investment and entrepreneurship to flourish.

In May 2020, in the middle of the first wave of the Covid-19 pandemic, Prime Minister Narendra Modi announced bold reforms in